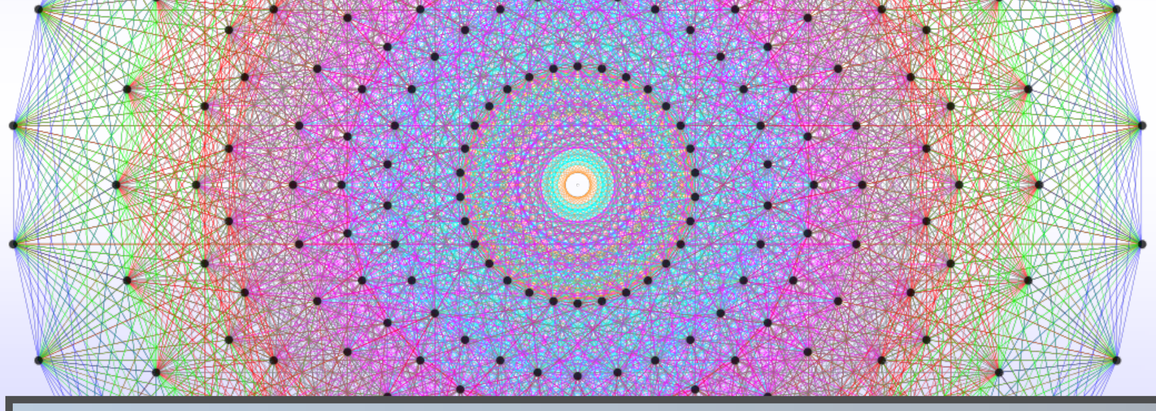


# E8

*The 2-D representation*

## How do the pieces fit?

Here we will collect the necessary resources to learn how any why connections are made in the 2 - D Exeptional Lie Group 8, and then we will use these resources to Code a representation of the Lie Group in C with the DISLIN Library.

## Lets Start with the Theory

### Brace Yourself

⇒ **Lie Group** A real Lie group is a group that is also a finite-dimensional real smooth manifold, in which the group operations of multiplication and inversion are smooth maps. Smoothness of the group multiplication, $\mu : G \times G \Rightarrow G, \mu(x, y) = xy$, means that is a smooth mapping of the product manifold G × G into G.
⇒ **E8** The Lie Group E8 has dimension 248, and its rank, the dimension of the maximal torus, is 8.
⇒ **Root System** We want to specify interesting root conditions of the above 8 - D system. Let E be a finite-dimensional Euclidean vector space, with the standard Euclidean inner product denoted by $(\cdot, \cdot)$. A root system $\Phi$ in E is a finite set of non-zero vectors (called roots) that satisfy the following conditions:

1. The roots span E.
2. The only scalar multiples of a root $\alpha \in \Phi$ that belong to $\Phi$ are $\pm\alpha$.
3. For every root $\alpha \in \Phi$ the set $\Phi$ is closed under reflection through the hyperplane perpendicular to $\alpha$.
4. (Integrality) If $\alpha$ and $\beta$ are roots in $\Phi$, then the projection of $\beta$ onto the line through $\alpha$ is an integer or half-integer multiple of $\alpha$.

### Into 2 - D

To go from the 8 dimensional space to the 2 - D projection of the object, we can project the root system in the Coxeter Plane.
⇒ **Coxeter Group** Formally, a Coxeter group can be defined as a group with the presentation $\langle r_1, ..r_n | (r_i, r_j)^m_{ij} = 1\rangle$ where $m_{ii} = 1$ and $m_{ij} \geq 2$ for $i \neq j$.
⇒ **Coexeter Element** A Coxeter element is a product of all simple reflections. The product depends on the order in which they are taken, but different orderings produce conjugate elements, which have the same order. The Coxeter number h is the order of a coxeter element.
⇒ **Coexeter Plane** For a given Coxeter element w, there is a unique plane P on which w acts by rotation by $2\pi/h$. This is called the Coxeter plane and is the plane on which P has eigenvalues $e^{2\pi i/h}$ and $e^{2\pi i/h} = e^{2\pi i(h1)/h}$.

### In English

We calculate the cross - section of E8 that we find most interesting, and project the result onto a plane.

## From Roots to Bytes: A matrix of Correspondence

### Points and lines

Let $B = \alpha_1, ..\alpha_n$ be a base of the root system.

$r_c = \Pi^n_{j=i} r_j$ where $r_j = Id - \frac{2\alpha_j \alpha_j^T}{(\alpha_j, \alpha_j)}$, where j = 1, .. n.

The Coxeter number is h, the smallest number so that $(r_c)^h = Id$.

We calculate the eigenvectors of the system as $r_c z = e^{2\pi i/h} z$.

The Coxeter plane is the space spanned by $x = Re(z)$ and $y = Im(z)$, $C = \Re x + \Re y$.

**Vertices** on the plane are given by $\alpha_x = (\alpha, x)$ and $\alpha_y = (\alpha, y)$.

**Edges** in the graph exist between roots $(\alpha, \beta)$ if the scalar product $(\alpha - \beta, \alpha - \beta)$ is minimised.

### Heuristics

Find a Root System of a Lie Group of order 8.

Run a minimising function on each of the 8 - D roots, pairwise, to find out if they are neighbours, (We expect that there are 56 neibours for each point, and that they are at distance $\sqrt{2}$) and produce a matrix of correspondence.

Create another point wise equivalent matrix for the position of each vertex on the coxeter plane.

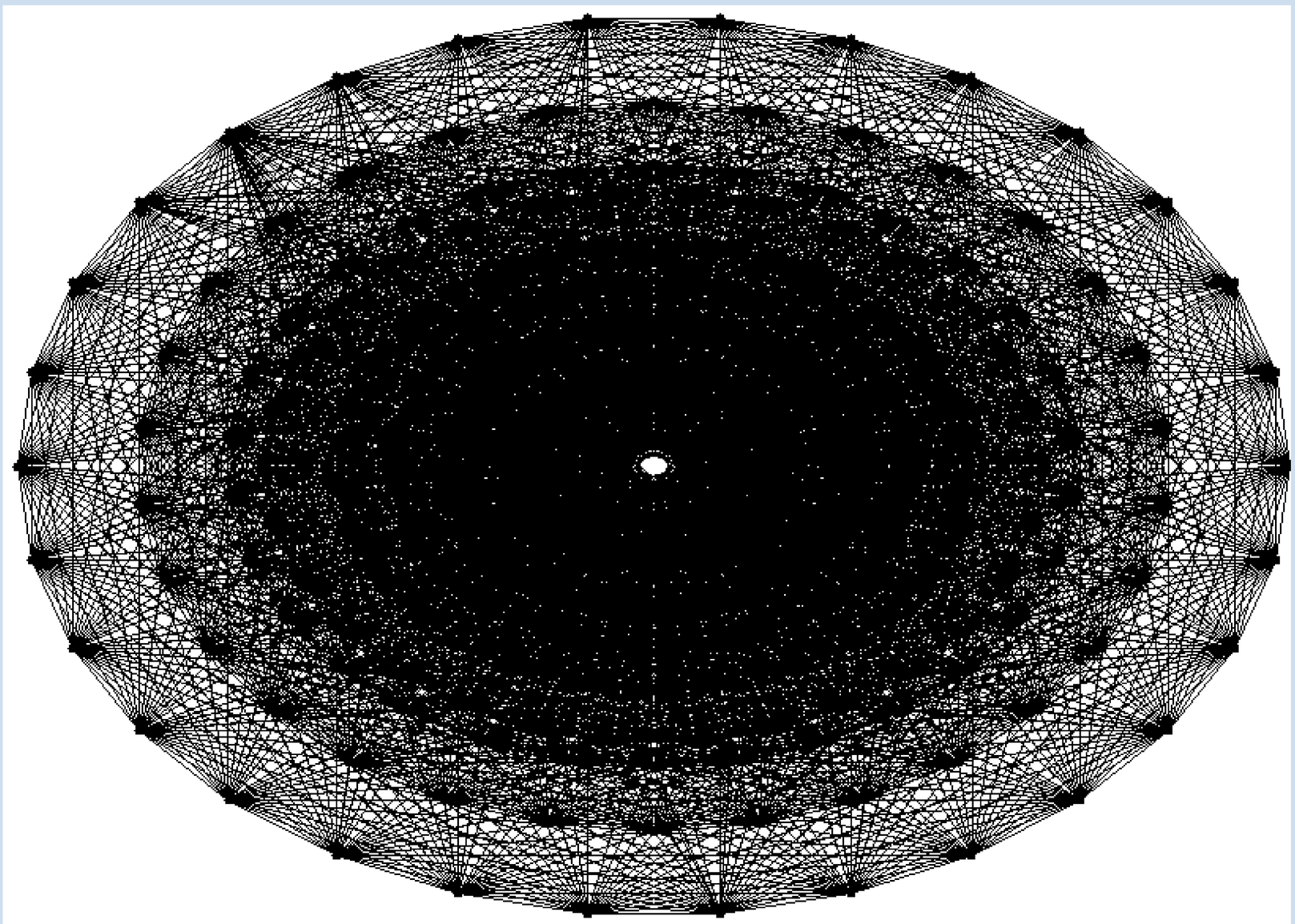
Use a graphical library to plot the graph.

## Code

**Repository for the code of this project**

https://github.com/stackedflows/C/tree/main/DISLIN

## Console Output



*The output image, known as the $4_{21}$ Gosset polytope has 240 vertices and 6720 edges, and*

## Resources


https://tamasgorbe.wordpress.com/2015/10/28/coxeter-projection-of-exceptional-root-systems/

http://www.madore.org/ david/math/e8rotate.html